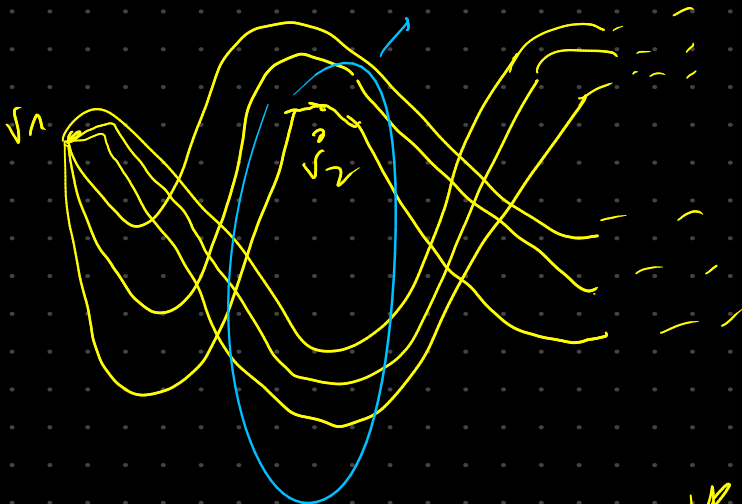


objectif: from special cases of H-T that change the η pick
quantity strategy

* Weak H-T: multiple connected components linked by vertex v_1 (with v_2 changes order for edges adjacent in v_1)

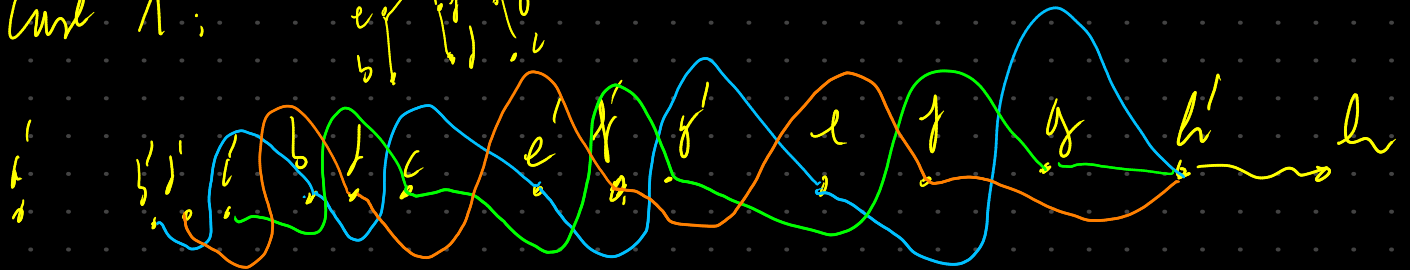
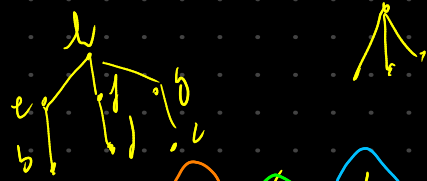


his case cannot happen if $h_{i,j}$ is connected



- Strong H-T: issues with odd crossings of adjacent edges.

Curve 1:



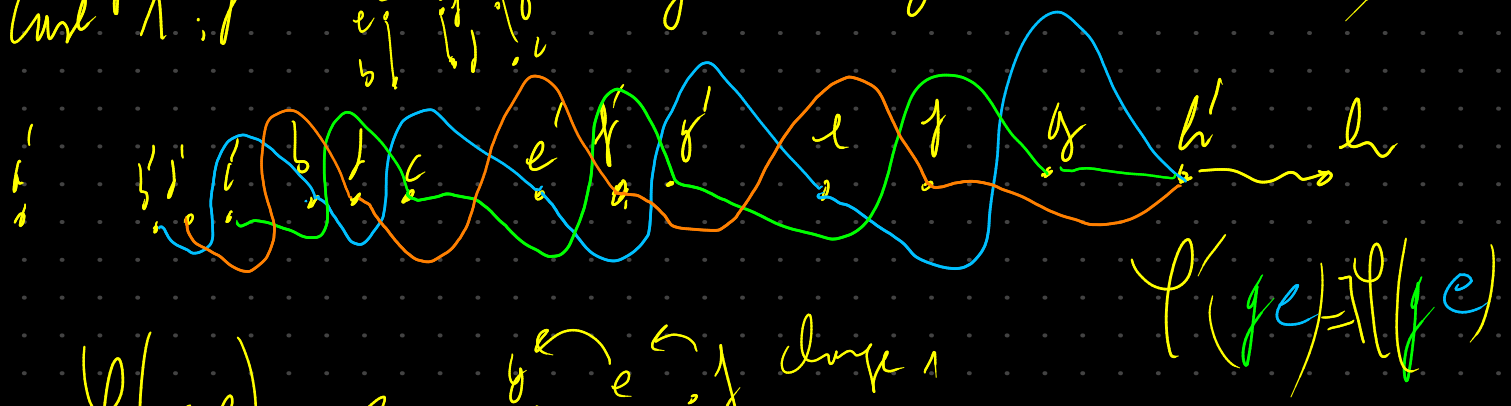
Cont 21



Cost {



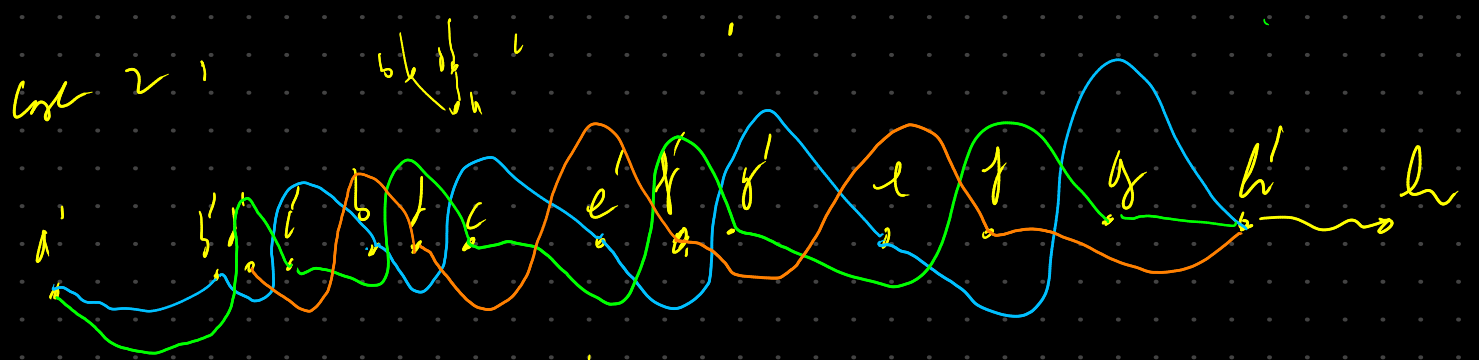
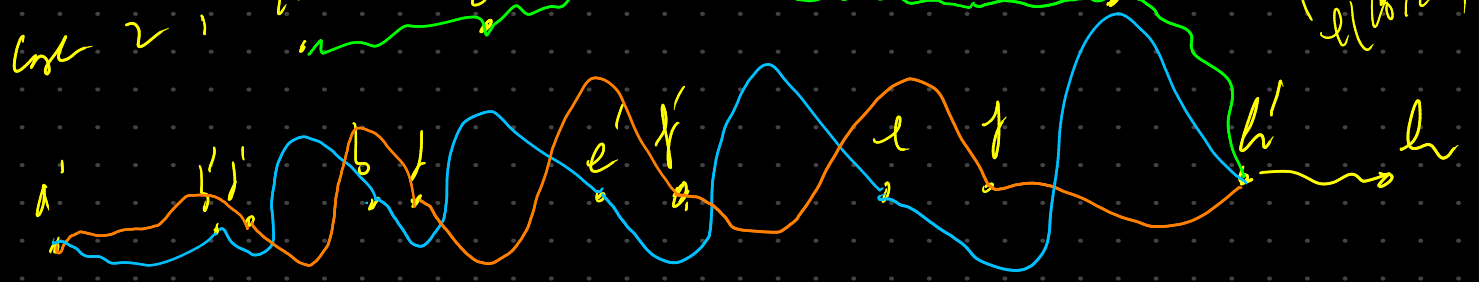
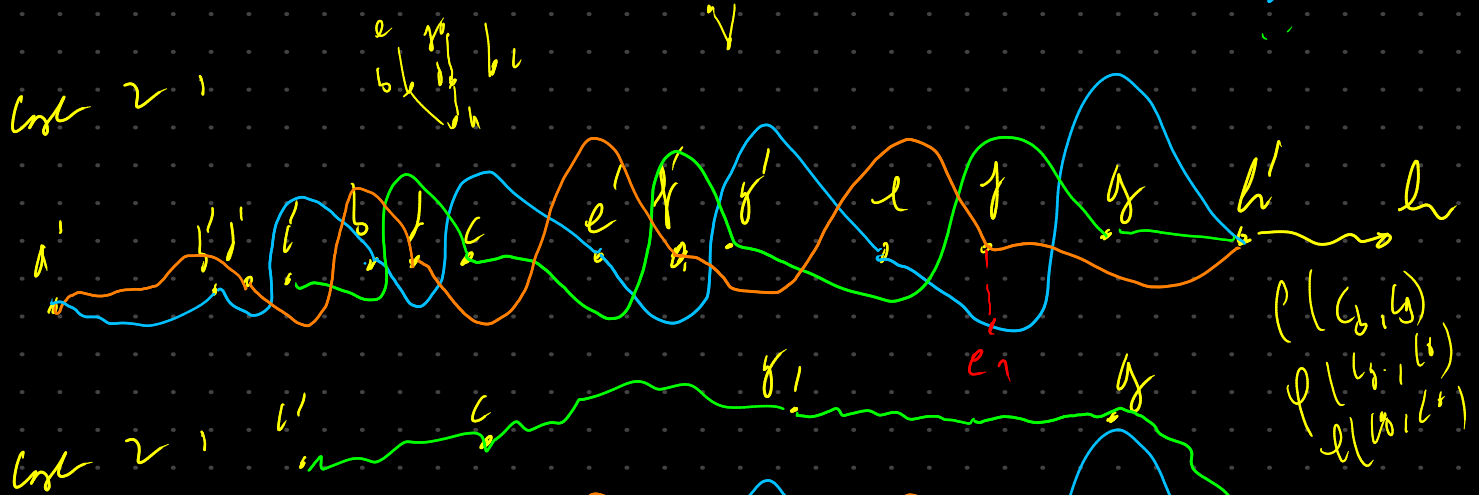
case 1: first odd crossing. (Strong H-T $\rightarrow$ weak H-T)



$P(ge) = 0$
 $P(gf) = 1$
 $P(ef) = 0$

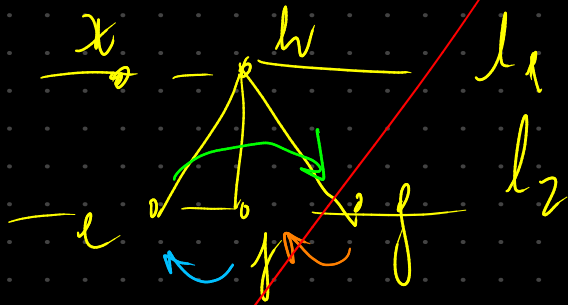
$P(g, L_g)$
 $P(L_g, L_o)$
 $P(L_o, L_g)$ no cycle

if we are in this situation



the one not in the cycle out.

let's consider a level planar graph h .



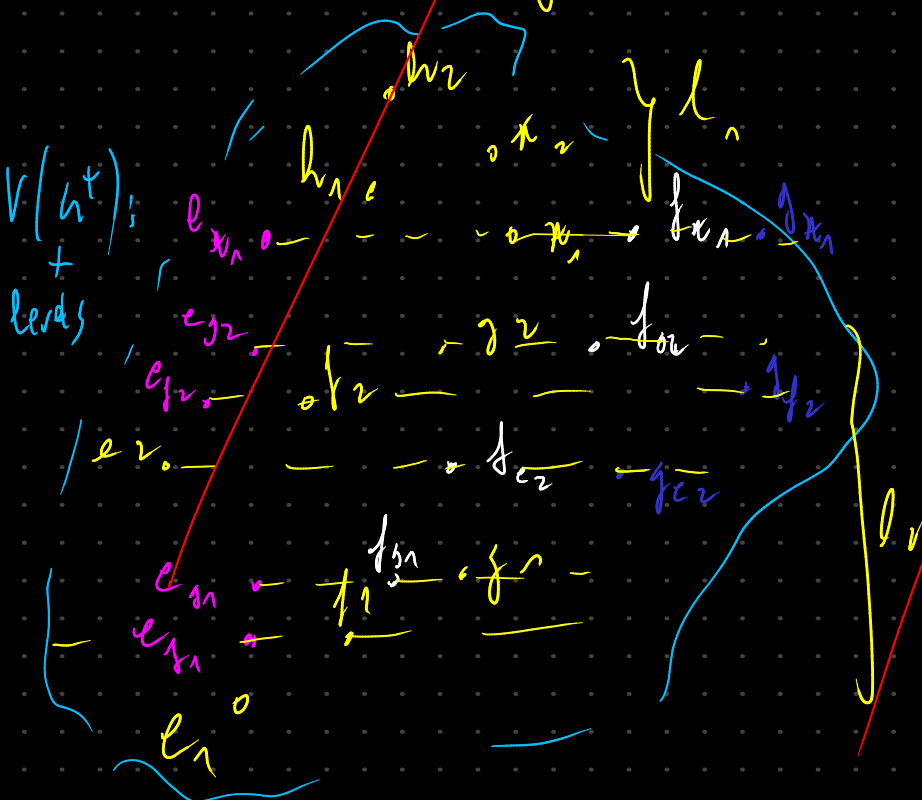
A satisfiable truth assignment φ of 2-SAT
 $\varphi(xh) = 1$
 $\varphi(e) = 0$
 $\varphi(fg) = 1$
 $\varphi(fg) = 0$

we want to draw a Hanani-tutte drawing h^* from $S(h^+)$.

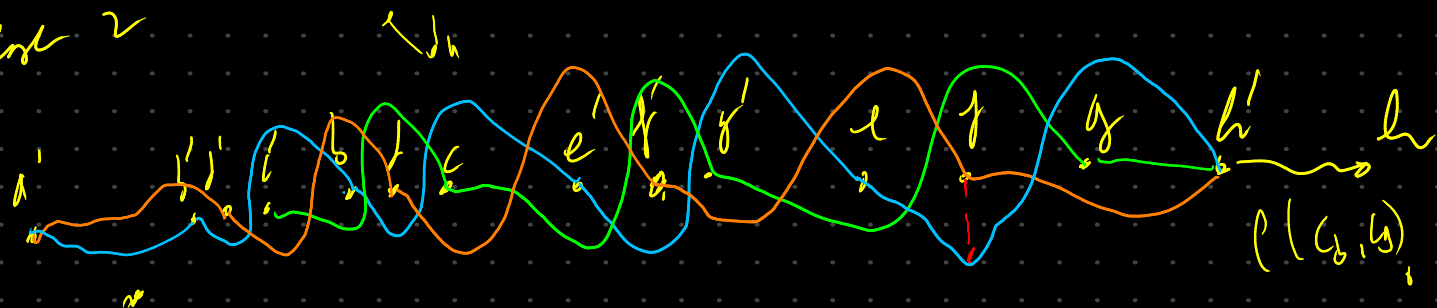
so first we formulate φ^+ of $S(h^+)$ from φ of $S(h)$
 following ^(prog form) Lemma 4 from level-planarity, transitivity
 vs. Even crossing paper, we have:

$$\varphi^+(e_{x_1}, f_{x_1}) = \varphi(o(e), o(f)) = 0$$

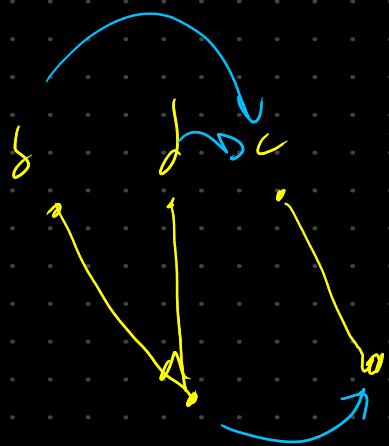
$$\varphi^+(e_{x_1}, f_{x_1}) = 0$$



Case 2



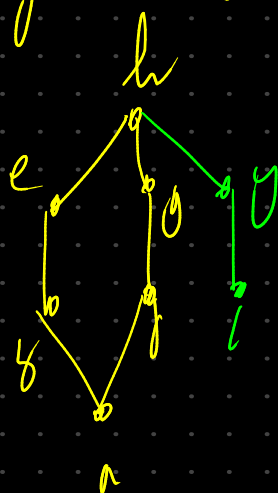
the green connected component can't have an edge from c' to its left in this drawing induced by φ^+ of $S(h^+)$. because:



it will break the cyclic ordering $\Rightarrow$

φ^+ is not a satisfying truth assignment to $S(h^+)$

so it doesn't matter if the green connected component is inside the cycle or not



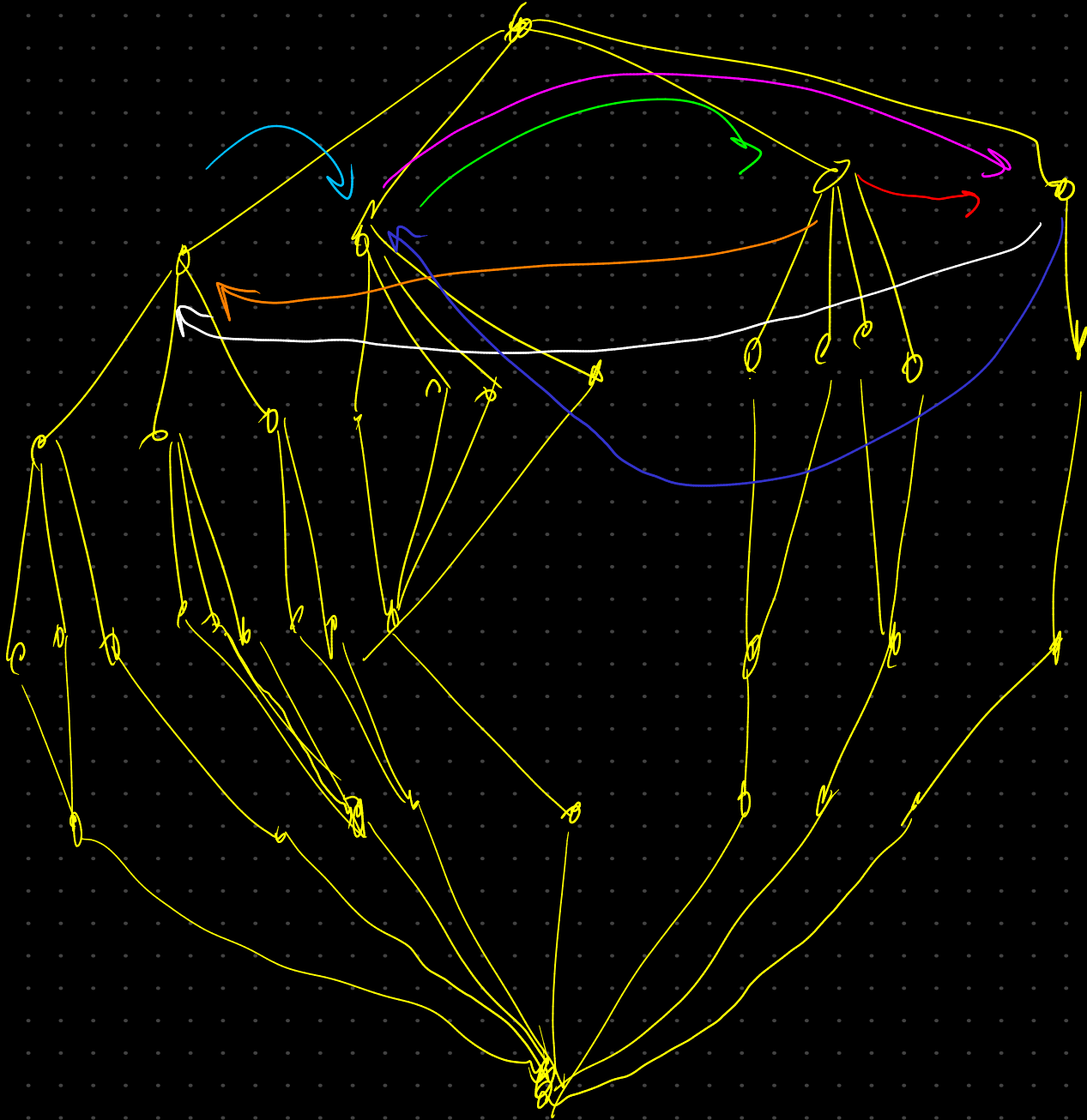
the green connected comp don't go beyond $L(C)$



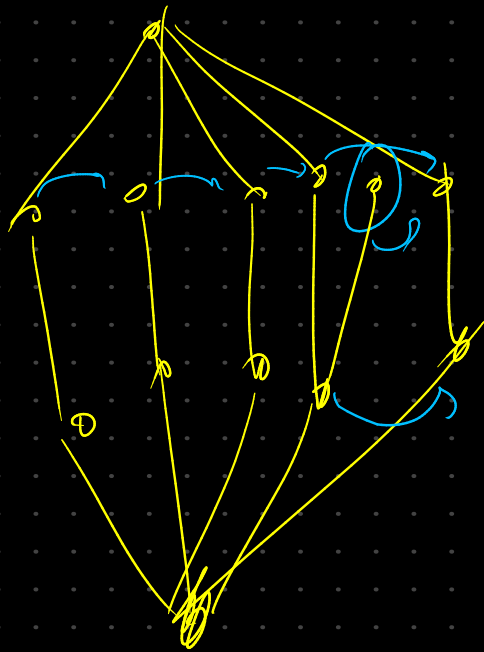
fixing a non-cyclic order between the disjoint vertices of adjacent edges will fix all the 3 cases.

→ weak H₁

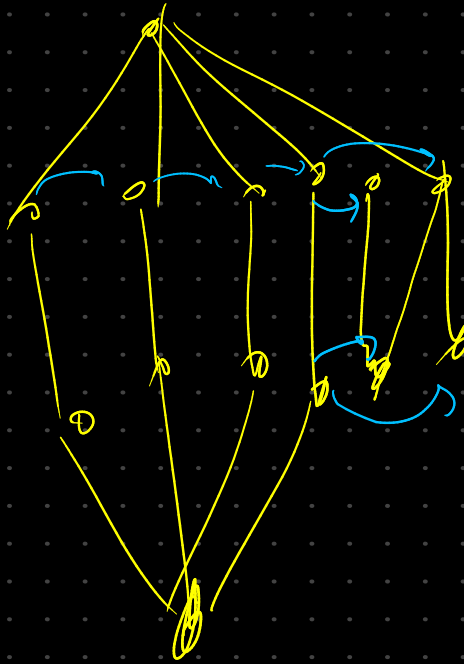
Case 3



then fix vertices that are in this situation



① ②



it takes out also
the issue
of the weak
H-T
and eq. drawing.